

Quiz (Carolina Reaper)

- Q1.** A shipping company has a quadratic cost function $C(x) = x^2 + 5x + 10$ and a linear revenue function $R(x) = 20x - 5$. How many units x must be sold to break even?
- (A) 2
(B) 5
(C) 10
(D) 12
(E) 15
- Q2.** Solve the system: $x^2 + 3y = 7$, $2x + y = 4$. What is the value of x ?
- (A) -1
(B) 1
(C) 2
(D) 3
(E) 4
- Q3.** A traffic flow model is given by $f(x) = x^2 - 4x + 5$ and $g(x) = 2x - 1$. Find the intersection point.
- (A) (1, 1)
(B) (2, 3)
(C) (3, 5)
(D) (4, 7)
(E) (5, 9)
- Q4.** A delivery company has a quadratic function $d(x) = x^2 - 6x + 9$ and a linear function $t(x) = 3x - 2$. Find the value of x where $d(x) = t(x)$.
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q5.** Solve the system: $y = x^2 - 2x - 3$, $y = 1 - x$. Find the value of y .
- (A) -2
(B) -1
(C) 0
(D) 1
(E) 2
- Q6.** A logistics company has a cost function $C(x) = x^2 + 2x + 1$ and a revenue function $R(x) = 4x - 3$. What is the break-even point?
- (A) (1, 1)
(B) (2, 5)
(C) (3, 6)
(D) (4, 13)
(E) (5, 16)
- Q7.** Solve the system: $x^2 + y = 4$, $x - y = 2$. Find the value of y .
- (A) -2
(B) -1
(C) 0
(D) 1
(E) 2
- Q8.** A transportation model is given by $f(x) = x^2 - x - 6$ and $g(x) = x + 2$. Find the intersection point.
- (A) (-2, 0)
(B) (1, 3)
(C) (2, 4)
(D) (3, 5)
(E) (4, 6)

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $y = x^2 + 2x - 6$, $y = -x + 4$. Find the intersection points and interpret the results in the context of a logistics company.
- Q10.** A shipping company has a quadratic cost function $C(x) = x^2 + 4x + 5$ and a linear revenue function $R(x) = 15x - 2$. Find the break-even point and interpret the results.

Chef's Tips

- Check for the correct interpretation of intersection points.
- Ensure students provide clear and concise explanations for their answers.
- Remind students to show all steps in their calculations.